

Inducible BRAF suppression models for melanoma tumorigenesis.

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Somatic mutations in BRAF have been reported in 50 to 70% of melanomas. The most common mutation is a valine to glutamic acid substitution at codon 600 (V600E). (V600E)BRAF constitutively activates ERK signaling and promotes proliferation, survival, and tumor growth. However, although BRAF is mutated in up to 80% of benign nevi, they rarely progress into melanoma. This implicates the BRAF mutation to be an initiating event that requires additional lesions in the genome for full-blown progression to melanoma. Even though the mutations appear early during the pathogenesis of melanoma, targeted BRAF knockdown using inducible shRNA in melanoma cell lines with BRAF mutations shows that BRAF is required for growth and maintenance of tumor in xenograft models.

BRAF V600E Expression in Primary Melanoma and Its Association With Death: A Population-Based, Retrospective, Cross-Sectional Study.

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Approximately 50% of melanomas contain BRAF mutations; the effects on survival are unclear. We aimed to determine whether mutant BRAF expression in melanoma differs according to age, sex, and melanoma-specific survival. A total of 638 patients who resided in Olmsted County, Minnesota, with a first lifetime diagnosis of melanoma between 1970 and 2009 were identified from the Rochester Epidemiology Project (REP). Available tissue was analyzed for a BRAF V600E mutation with immunohistochemistry.

Current Controversies and Challenges on BRAF V600K-Mutant Cutaneous Melanoma.

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About 50% of melanomas harbour a BRAF mutation. Of these 50%, 10% have a V600K mutation. Although it is the second most common driver mutation after V600E, no specific studies have been conducted to identify a clinical and therapeutic gold standard for this patient subgroup. We analysed articles, including registrative clinical trials, to identify common clinical and biological traits of the V600K melanoma population, including different adopted therapeutic strategies. Melanoma V600K seems to be more frequent in Caucasian, male and elderly populations with a history of chronic sun damage and exposure. Prognosis is poor and no specific prognostic factor has been identified. Recent findings have underlined how melanoma V600K seems to be less dependent on the ERK/MAPK pathway, with a higher expression of PI3KB and a strong inhibition of multiple antiapoptotic pathways. Both target therapy with BRAF inhibitors + MEK inhibitors and immunotherapy with anti-checkpoint blockades are effective in melanoma V600K, although no sufficient evidence can currently support a formal recommendation for first line treatment choice in IIIC unresectable/IV stage patients. Still, melanoma V600K represents an unmet medical need and a marker of poor prognosis for cutaneous melanoma.

BRAF inhibitor activity in V600R metastatic melanoma.

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Activating mutations in the BRAF gene occur in approximately 50% of melanomas. More than 70% of BRAF mutations are V600E and 10-30% are V600K. Potent and selective BRAF inhibitors have demonstrated significant clinical benefits in patients with V600E and V600K BRAF-mutated melanoma. V600R mutations constitute approximately 3-7% of all BRAF mutations and the activity of BRAF inhibitors in patients with this mutation is unknown. We have treated 45 patients with V600 mutated melanoma including patients with V600R mutation between July 2011 and October 2012 with the selective BRAF inhibitor dabrafenib (n=43) or vemurafenib (n=2) via a compassionate access programme. The overall response rate was 50% for the whole population with a progression-free survival of 5.5 months. Five objective responses were seen in six assessable patients with V600R BRAF mutation (n=9). Our experience suggests that patients with V600R BRAF mutations can be treated successfully with oral BRAF inhibitors, and molecular diagnostic assays should include detection of this type of mutation.

BRAFV600E mutations in malignant melanoma are associated with increased expressions of BAALC.

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BACHGROUND: Activating BRAF mutations are present in approximately 50% of melanomas. Although different downstream target genes of the most common mutant V600E have been identified, the contribution of activating BRAF mutations to malignant transformation needs further clarification. Microarray gene analysis was performed for human melanoma cell lines harboring BRAFV600E mutations in comparison to cell lines without this mutation. This analysis revealed a more than two fold down-regulation of 43 and an increase of 39 gene products. BAALC (Brain and acute Leukaemia, cytoplasmatic) was most prominently regulated, since it was up-regulated in mutated cell lines by a mean of 11.45. Real time PCR analyses with RNA from melanoma cell lines (n = 30) confirmed the BRAF-activation dependent up-regulation of BAALC. BAALC, which has been associated with cell dedifferentiation and migration, may function as a downstream effector of activating BRAF mutations during melanomagenesis.

[BRAF mutation testing for the choice of melanoma treatment].

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Individual life-time risk of melanoma in white residents of highly developed countries may be as high as 2%. Continuing rise of melanoma incidence is directly related to the improving life standards, especially to growing opportunities of attending sea resorts and getting tanned. Melanoma is usually highly aggressive and resistant to standard cytotoxic therapy, therefore 5-years survival of patients with the metastatic disease does not exceed 10-15%. Approximately 50% of melanomas contain point mutation in codon 600 of the BRAF kinase. Specific inhibitors of activated BRAF have demonstrated unprecedented therapeutic efficacy, thus BRAF testing has become a mandatory component of treatment planning for inoperable melanoma. This review discusses key issues, which are related to various clinical, morphological, and molecular genetic aspects of determination of BRAF status in metastatic melanoma.

BRAF inhibitor rechallenge in patients with advanced BRAF V600-mutant melanoma.

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In around 50% of melanomas, the BRAF V600 mutation, resulting in an activation of the MAP kinase pathway, is detected. BRAF inhibitors have shown remarkable activity on the disease. However, efficacy is short-lived in most cases, with a median disease-free survival of 6 months. This short duration of response could be explained by the acquisition of resistance mechanisms. Some cancers show sensitivity to the reintroduction of previously active drugs after disease progression. We carried out a retrospective monocentric study on patients with BRAF V600-mutated melanoma who were rechallenged with BRAF inhibitors that were previously beneficial, but in whom the disease had progressed. Nine patients were included. Five patients showed a subsequent partial response, two showed a dissociated response leading to clinical improvement, and two showed no radiological nor clinical response. Eight patients who received rechallenge BRAF inhibitor had received an intercurrent treatment with ipilimumab. These cases suggest that intermittent treatment with BRAF inhibitors could provide clinical benefit and that sequential therapies should be further evaluated in clinical trials.

Natural killer cells are essential for the ability of BRAF inhibitors to control BRAFV600E-mutant metastatic melanoma.

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BRAF(V600E) is a major oncogenic mutation found in approximately 50% of human melanoma that confers constitutive activation of the MAPK pathway and increased melanoma growth. Inhibition of BRAF(V600E) by oncogene targeting therapy increases overall survival of patients with melanoma, but is unable to produce many durable responses. Adaptive drug resistance remains the main limitation to BRAF(V600E) inhibitor clinical efficacy and immune-based strategies could be useful to overcome disease relapse. Tumor microenvironment greatly differs between visceral metastasis and primary cutaneous melanoma, and the mechanisms involved in the antimetastatic efficacy of BRAF(V600E) inhibitors remain to be determined. To address this question, we developed a metastatic BRAF(V600E)-mutant melanoma cell line and demonstrated that the antimetastatic properties of BRAF inhibitor PLX4720 (a research analogue of vemurafenib) require host natural killer (NK) cells and perforin. Indeed, PLX4720 not only directly limited BRAF(V600E)-induced tumor cell proliferation, but also affected NK cell functions. We showed that PLX4720 increases the phosphorylation of ERK1/2, CD69 expression, and proliferation of mouse NK cells in vitro. NK cell frequencies were significantly enhanced by PLX4720 specifically in the lungs of mice with BRAF(V600E) lung metastases. Furthermore, PLX4720 also increased human NK cell pERK1/2, CD69 expression, and IFN γ release in the context of anti-NKp30 and IL2 stimulation. Overall, this study supports the idea that additional NK cell-based immunotherapy (by checkpoint blockade or agonists or cytokines) may combine well with BRAF(V600E) inhibitor therapy to promote more durable responses in melanoma.

Mycobacterium bovis BCG in metastatic melanoma therapy.

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Melanoma is the most aggressive form of skin cancer, with a high mortality rate and with 96,480 new

cases expected in 2019 in the US. BRAFV600E, the most common driver mutation, is found in around 50% of melanomas, contributing to tumor growth, angiogenesis, and metastatic progression. Dacarbazine (DTIC), an alkylating agent, was the first chemotherapeutic agent approved by the US Food and Drug Administration (FDA) used as a standard treatment. Since then, immunotherapies have been approved for metastatic melanoma (MM) including ipilimumab and pembrolizumab checkpoint inhibitors that help decrease the risk of progression. Moreover, *Mycobacterium bovis* Bacillus Calmette-Guerin (BCG) serves as an adjuvant therapy that induces the recruitment of natural killer NK, CD4+, and CD8+ T cells and contributes to antitumor immunity. BCG can be administered in combination with chemotherapeutic and immunotherapeutic agents and can be genetically manipulated to produce recombinant BCG (rBCG) strains that express heterologous proteins or overexpress immunogenic proteins, increasing the immune response and improving patient survival. In this review, we highlight several studies utilizing rBCG immunotherapy for MM in combination with other therapeutic agents.

The Prognostic Value of miR-125b, miR-200c and miR-205 in Primary Cutaneous Malignant Melanoma Is Independent of BRAF Mutational Status.

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BRAF mutations are present in around 50% of cutaneous malignant melanomas and are related to a poor outcome in advanced-stage melanoma patients. miRNAs are epigenetic regulators that modulate different cellular processes in cancer, including melanoma development and progression. However, there are no studies on the potential associations of the genetic alterations of the BRAF gene with miRNA expression in primary cutaneous melanomas. Here, in order to analyze the influence of BRAF mutations in the ability of selected miRNAs to predict clinical outcome and patient survival at the time of diagnosis, we studied the prognostic value of miR-125b, miR-200c and miR-205 expression depending on the BRAF mutational status in fresh, frozen primary tumor specimens. For this purpose, RNA was extracted for studying both BRAF mutations by Sanger sequencing and miRNA expression. Our results indicate that, although there seems to be a slight preference for their predictive ability in the BRAF mutated group, the expression of these three miRNAs serves effectively to predict the clinical outcome of melanoma patients independently of BRAF mutational status at the time of primary tumor diagnosis.
